Statistical evidence for golden ratio structure in elementary fermion mass ratios with QCD regime dependence

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Abstract

We report a statistically significant golden ratio ($\Phi = (1+\sqrt{5})/2 \approx 1.618$) structure in the mass ratios of elementary fermions. Starting from the electron mass as the sole seed of a recursive bootstrap algorithm, 11 Standard Model particle masses are reproduced using five Φ-based operators, with a mean error of 2.5%. The operator exponents $\{3, 4, 11\}$ are Lucas numbers (L_2, L_3, L_5; the selection of these three from the infinite Lucas sequence is empirical), and the non-integer operators Φ^Φ and $\Phi^{1/\Phi}$ satisfy $\Phi^\Phi/\Phi^{1/\Phi} = \Phi$ (proven from $\Phi^2 = \Phi + 1$). A universality test shows no alternative constant $K \in [1.2, 2.5]$ describes all five core relations simultaneously with integer exponents ($p = 0.001$). A basis scan over 500 alternatives confirms that among standard mathematical constants, only Φ achieves full reproduction at 10% tolerance (e: 1/11, π: 1/11, base 2: 9/11); all successful bases in the scan are powers of Φ. The six independent Φ-ladders defined by nine fermion masses are significantly uniformly distributed (Greenwood test, $p = 0.007$). Chain connectivity at 5% tolerance is highly significant ($p = 0.006$) and strengthens at tighter tolerance. In the hadron spectrum, Φ-relations manifest exclusively in the QCD-dominated regime (binding energy $> 50\%$ of hadron mass): Fisher exact test yields $p = 0.008$ with perfect separation (4/4 vs. 0/5). A holdout test over 167 PDG hadrons finds no statistically significant Φ-grid proximity ($p = 0.055$, one-sided; $p = 0.91$ with conservative 4-seed subset), and gauge bosons show no Φ-signal (all $p > 0.5$). Three specific hadron relations in the light QCD regime survive all controls (Fisher combined $p = 2.2 \times 10^{-5}$). All results are reproducible via companion code.

Keywords: fermion masses, golden ratio, mass relations, Lucas numbers, QCD regime dependence, hadron spectrum, statistical test

1 Introduction

The Standard Model of particle physics contains nine Yukawa coupling constants that determine the masses of the charged fermions (6 quarks and 3 charged leptons); with massive neutrinos, additional parameters enter. None of these are predicted by the theory; they are extracted from experiment and span more than five orders of magnitude, from $m_e = 0.511$ MeV to $m_t = 172{,}760$ MeV. Whether an organizing principle underlies these apparently arbitrary values remains one of the central open questions in particle physics [22, 23].

Several empirical mass relations have been proposed. Nambu [8] observed that particle masses are approximately quantized in units of $m_e/\alpha \approx 70$ MeV, where α is the fine structure constant. Barut [9] proposed a lepton mass formula based on $\alpha^{-1}n^4$ scaling. MacGregor [10] systematized

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these observations as "α-quantization." The most prominent is the Koide formula [1], which relates the three charged lepton masses through the surprisingly precise identity

$$\frac{m_e + m_\mu + m_\tau}{\left(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau}\right)^2} = \frac{2}{3}\,, \tag{1}$$

satisfied to better than 0.01%. Despite extensive theoretical work [4, 3, 5, 6, 7], the physical origin of this relation remains unexplained. Extensions to the quark sector show approximate validity [3].

In the present work, we report a different class of mass relations based on the golden ratio $\Phi = (1+\sqrt{5})/2 \approx 1.618$. The golden ratio has appeared in condensed matter physics as an emergent symmetry ratio in quantum critical Ising chains [20]; its appearance in the particle mass spectrum has not been systematically investigated before. We perform a systematic statistical analysis, including falsification tests, negative controls, and a regime-dependent analysis of the hadron spectrum.

We emphasize that this paper reports an empirical observation with rigorous statistical evaluation, not a new theory. We do not claim to explain *why* Φ appears; we demonstrate *that* it appears, quantify the statistical significance, delineate the domain of validity, and present falsifiable predictions.

2 The Φ-chain: fermion masses from the electron

2.1 Mass spectrum

Starting from the electron mass $m_e = 0.511$ MeV as the sole seed, we construct a correlation graph in which each particle mass is related to a "parent" mass by one of five Φ-operators or by a geometric mean (impedance) of two already-identified masses. Each prediction in the table below uses the PDG mass of the parent particle, not a chain-propagated value; the algorithm is defined in Appendix A.

Table 1 shows the resulting predictions against PDG 2024 reference values [11]. Each row represents a single algebraic relation: the predicted mass is computed from the PDG value of the parent particle.

Table 1: Genesis predictions vs. PDG 2024 reference masses. Each prediction uses the PDG mass of the parent particle. Rows are sorted by mass, not by algorithmic generation order; see Appendix A for the iterative construction. $\Phi = (1+\sqrt{5})/2 = 1.6180339887\ldots$

Particle	M_{pred} (MeV)	M_{PDG} (MeV)	Relation	Error
Electron	0.511	0.511	Seed	—
Up	2.165	2.16	$m_e \times \Phi^3$	0.2%
Down	4.706	4.67	$m_u \times \Phi^\Phi$	0.8%
Strange	95.02	93.4	$\sqrt{m_u \cdot m_b}$	1.7%
Muon	101.69	105.66	$m_e \times \Phi^{11}$	3.8%
Pion	139.72	139.57	$\sqrt{m_d \cdot m_b}$	0.1%
Proton	956.6	938.3	$m_\pi \times \Phi^4$	2.0%
Charm	1,263	1,270	$m_p \times \Phi^{1/\Phi}$	0.5%
Tau	1,710	1,777	$m_c \times \Phi^{1/\Phi}$	3.8%
Bottom	4,017	4,180	$\sqrt{m_s \cdot m_t}$	3.9%
Top	186,721	172,760	$m_p \times \Phi^{11}$	8.1%
Mean error (10 non-seed particles)				2.5%
Median error				1.8%

Figure 1 shows the predictions on a log–log scale.

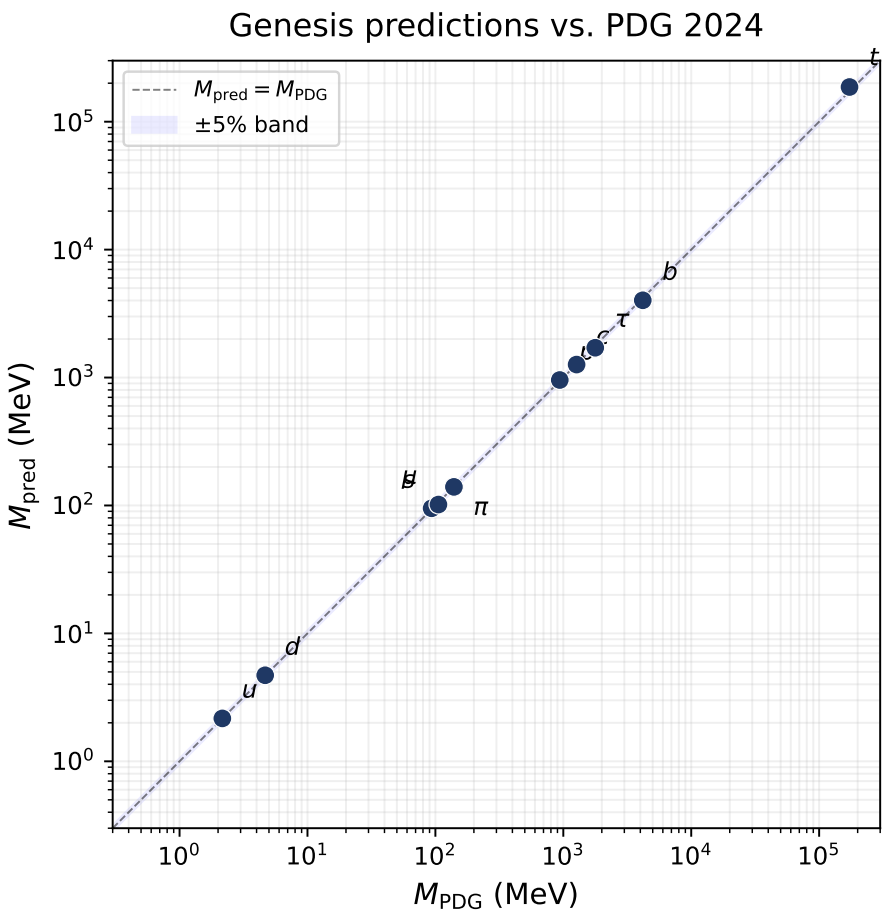

Figure 1: Genesis predictions vs. PDG 2024 masses on a log–log scale. The dashed line indicates perfect agreement; the shaded band shows ±5%. All 10 non-seed particles fall within 8.1% of the diagonal.

Note on mass conventions. The reference masses follow the PDG standard [11]: pole masses for leptons and the top quark,[1] $\overline{\text{MS}}$ at $\mu = 2$ GeV for light quarks (u, d, s), $\overline{\text{MS}}$ at $\mu = m_q$ for charm and bottom, and physical masses for pion and proton. Four of the eleven chain entries depend on this convention (those using u, d, or s as parent mass); a convention-independent subset is tested separately in Section 3.4. The bottom quark ($\overline{\text{MS}}$ at $\mu = m_b$) is counted as convention-independent because the self-scale running mass is uniquely defined and lies well within the perturbative QCD regime ($m_b \gg \Lambda_{\text{QCD}}$), whereas the light quark masses at $\mu = 2$ GeV depend on an external choice of renormalization point.

2.2 Algebraic structure of the operators

The five operators $\{\Phi^3, \Phi^4, \Phi^{11}, \Phi^{\Phi}, \Phi^{1/\Phi}\}$ possess nontrivial algebraic properties:

Non-integer operators. The identity

$$\frac{\Phi^{\Phi}}{\Phi^{1/\Phi}} = \Phi^{\,\Phi - 1/\Phi} = \Phi^1 = \Phi \tag{2}$$

[1] We use $m_t = 172{,}760$ MeV following the PDG 2020 world average. The PDG 2024 update gives $172{,}570 \pm 290$ MeV; substituting this value shifts the mean Genesis error by less than 0.02 percentage points and leaves all statistical conclusions unchanged.

follows directly from $\Phi^2 = \Phi + 1$:

$$\Phi - \frac{1}{\Phi} = \frac{\Phi^2 - 1}{\Phi} = \frac{(\Phi + 1) - 1}{\Phi} = 1\,. \tag{3}$$

This is an exact algebraic identity, not a numerical approximation. The two non-integer operators contribute zero free parameters.

Integer exponents and Lucas numbers. The exponents $\{3, 4, 11\}$ are Lucas numbers [21]: $L_2 = 3$, $L_3 = 4$, $L_5 = 11$, where the Lucas sequence is defined by $L_0 = 2$, $L_1 = 1$, $L_n = L_{n-1} + L_{n-2}$. (Note: these are not consecutive Lucas numbers; $L_4 = 7$ is absent.) Lucas numbers satisfy $\mathrm{round}(\Phi^n) = L_n$ for $n \geq 2$, i.e., they are the nearest-integer approximation of Φ-powers. Notably, 4 and 11 are *not* Fibonacci numbers; only the Lucas sequence contains all three exponents.

Free parameters. Given the algebraic constraints above, the five operators contain zero free exponents. *Qualification:* structural degrees of freedom remain — specifically the target set, mass definitions, the 20% matching tolerance, and the search rules. These are documented in Appendix A.

3 Statistical validation

3.1 Φ-universality test

Question: Is there an alternative constant $K \neq \Phi$ that simultaneously describes all core relations with integer exponents?

For each of five independent mass ratios (Table 2), we compute the Φ-exponent $n = \log(\mathrm{ratio})/\log(\Phi)$ and its distance to the nearest integer, $\Delta n = |n - \mathrm{round}(n)|$.

Table 2: Five core mass relations and their Φ-exponents.

Relation	Mass ratio	n_Φ	round(n)	Δn
m_u/m_e	4.227	2.996	3	0.004
m_μ/m_e	206.77	11.080	11	0.080
m_τ/m_e	3,477.2	16.945	17	0.055
M_{Ω^-}/m_s	17.906	5.996	6	0.004
$\Delta M_{\mathrm{dec}}/\Delta m_q$	1.632	1.017	1	0.017

The observed maximum is $\max|\Delta n|(\Phi) = 0.080$ (muon–electron ratio).

Monte Carlo test: We draw 2×10^5 random constants K uniformly in $[1.2, 2.5]$. For each K, we compute $\max|\Delta n|(K)$ over all five relations.

Result: $P\big(\max|\Delta n|(K) \leq 0.080\big) = 0.0009$. Fewer than 1‰ of constants describe all five relations as well as Φ (Figure 2).

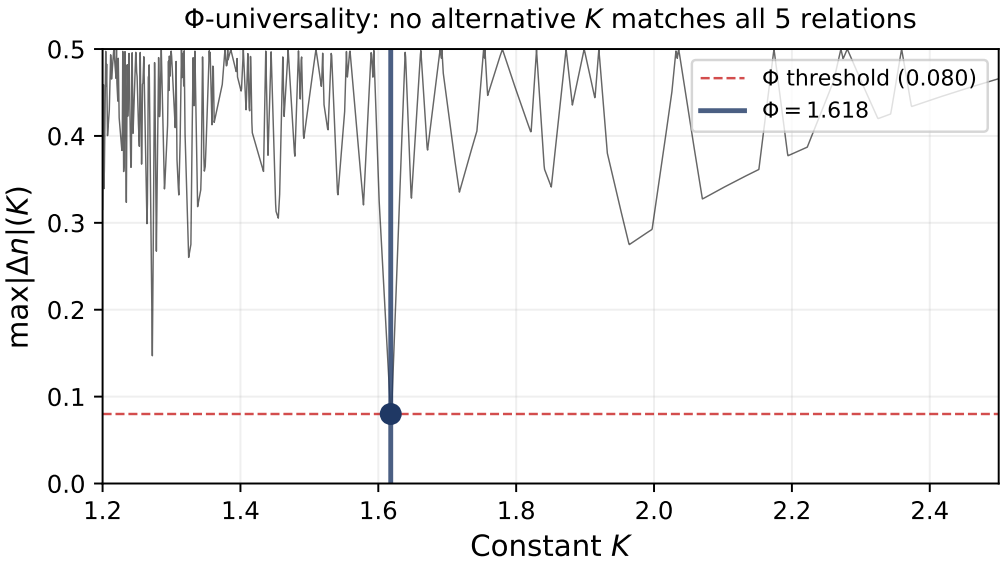

Figure 2: Φ-universality test: the maximum deviation $\max|\Delta n|(K)$ over five core mass relations, scanned across constants $K \in [1.2, 2.5]$. The golden ratio (vertical line, $\Phi = 1.618$) achieves the joint minimum. No other constant below the dashed threshold exists in the scanned range.

3.2 Phase uniformity of Φ-ladders

Each fermion mass m defines a Φ-phase $\phi = \{\log(m/m_e)/\log\Phi\} \in [0, 1)$, where $\{x\}$ denotes the fractional part. Particles with similar phase ($|\Delta\phi| < 0.08$, circular distance) belong to the same Φ-ladder.

The nine elementary fermion masses form six independent ladders (Table 3). Their phases are tested for uniformity on the circle $[0, 1)$ using the Greenwood statistic $G = \sum_i g_i^2$, where g_i are the circular gaps between adjacent ladder phases [17, 19].

Table 3: Six Φ-ladders from nine elementary fermion masses.

Ladder	Members	Phase (circular mean)
1	e, u, μ, τ	0.005
2	c	0.247
3	t	0.456
4	d	0.598
5	b	0.722
6	s	0.823

Result: $G_{\text{obs}} = 0.181$ vs. $G_{\min} = 1/6 = 0.167$ (perfect uniformity). Monte Carlo (5×10^5 random configurations of 6 points on $[0, 1)$): $P(G \leq G_{\text{obs}}) = 0.007$.

The six ladders are distributed more uniformly than 99.3% of random configurations. The Φ-lattice tiles the logarithmic mass space systematically, not randomly.

3.3 Chain connectivity

Question: How unusual is the real particle spectrum's connectivity under Φ-operators?

Starting from the seed (electron), we attempt to reach all 11 target masses through single operator applications or pairwise geometric means (impedances). We compare against 5,000 pseudo-spectra (11 masses drawn log-uniformly in $[m_e, m_t]$); results are shown in Table 4.

Table 4: Chain connectivity: real spectrum vs. pseudo-spectra.

Tolerance	Real (Φ)	Pseudo (mean)	p-value
10%	11/11	3.8/11	0.066
5%	**10/11**	1.8/11	**0.006**
3%	**7/11**	1.4/11	**0.005**

The signal *strengthens* at tighter tolerance — the opposite of a geometric coverage artifact. At 10% tolerance, random spectra occasionally achieve connectivity by chance. At 3%, chance connections vanish but the real signal persists. This is the qualitative fingerprint of structure versus noise.

3.4 Convention-independent subset

Seven of the eleven particles use only pole masses or physical masses that do not depend on a renormalization convention: e, μ, τ, pion, proton, bottom ($\overline{\text{MS}}$ at m_b, i.e., self-scale), and top.

For this seven-particle subset: chain connectivity at 5% yields $p = 0.019$; a basis scan over 300 random bases $\beta \in [1.05, 3.5]$ gives Φ rank #1/301 ($p = 0.003$). The signal exists independently of the 2 GeV convention.

3.5 Robustness: alternative bases and tolerances

A systematic scan over 500 equally spaced bases $\beta \in [1.05, 3.5]$ shows that 56 bases (11.2%) reproduce all 11 particles at 20% tolerance. These bases cluster in five groups, all centered at powers of Φ: near $\Phi \approx 1.62$ (12 bases), $\Phi^{1.4} \approx 2.0$ (15 bases), $\Phi^{1.9} \approx 2.5$ (7 bases), $\Phi^{2.4} \approx 3.2$ (20 bases), and $\Phi^{0.7} \approx 1.4$ (2 bases). No non-Φ-related region achieves full reproduction. At 10% tolerance, only two clusters survive: the Φ-neighborhood (4 bases) and its reparametrization $\Phi^{2.4}$ (6 bases); at 5%, no base achieves 11/11.

Crucially, the Φ-cluster separates from all alternatives at reduced tolerance (Table 5):

Table 5: Number of particles reproduced (out of 11) by the Genesis algorithm for different bases and tolerances.

Tolerance	Φ (1.618)	e (2.718)	π (3.142)	2
5%	10/11	1/11	1/11	1/11
10%	**11/11**	1/11	1/11	9/11
15%	11/11	1/11	10/11	11/11
20%	11/11	1/11	11/11	11/11

At 10% tolerance, Φ is the *only* standard mathematical constant that achieves complete reproduction. The Euler number e never finds more than the seed electron; π reaches only 1/11 at 10% but achieves 10/11 at 15% through the $\Phi^{2.4}$ reparametrization channel (since $\pi \approx 3.14$ lies near $\Phi^{2.4} \approx 3.17$). The base 2 requires $\geq 10\%$ tolerance (9/11) and reaches 11/11 at 15%. The secondary cluster at $\beta \approx 3.17 \approx \Phi^{2.4}$ that also achieves 11/11 at 10% is a reparametrization, not an independent constant: it exists because $\beta^{11} \approx m_t/m_e$, exploiting the same Φ-structured mass ratios through rescaled exponents. This asymmetry is the strongest single argument against a "numerology" interpretation: it shows that Φ is not interchangeable with other irrational constants.

3.6 PDG measurement uncertainties

PDG measurement uncertainties range from $< 10^{-6}$ (electron, muon) to $\sim 20\%$ (up quark). For 7 of 11 particles, the relative experimental uncertainty is below 1%, i.e., one to two orders of magnitude smaller than the typical Genesis deviation (2.5% mean). The three light quarks (u, d, s) and charm carry uncertainties of order 1–20%. Sensitivity to PDG measurement uncertainties varies across tests and is governed by which input masses carry large relative errors. The lepton masses (m_e, m_μ, m_τ) are known to $< 10^{-5}$ relative precision, so any test depending only on leptons is insensitive to measurement uncertainties. Tests involving light quark masses, however, inherit the large $\overline{\text{MS}}$ uncertainties: $m_u = 2.16^{+0.49}_{-0.26}$ MeV ($\sim 20\%$) and $m_s = 93.4^{+8.6}_{-3.4}$ MeV ($\sim 6\%$). A systematic $\pm 1\sigma$ perturbation analysis (5×10^5 Monte Carlo trials per configuration; code in companion repository) shows the following:

- **Robust** (p-value shift < 0.01): $m_\mu/m_e \approx \Phi^{11}$ and $m_\tau/m_e \approx \Phi^{17}$ (lepton sub-chain), A4 regime hypothesis (binary classification insensitive to quark mass shifts), B3 b-replacement (depends only on τ and directly measured hadron masses).
- **Moderately sensitive** (p-value shift $\lesssim 0.1$): A2 Greenwood test (cluster composition and gap structure shift when m_s moves by 1σ, potentially changing the number of effective ladders), B1 Ω^- prediction (degraded by factor 42 in Δn at $m_s + 1\sigma$).
- **Strongly sensitive**: A1 Genesis chain ($m_u/m_e \approx \Phi^3$ degrades from $\Delta n = 0.004$ to 0.42 at $m_u + 1\sigma$), A3 Φ-universality (p shifts from 0.001 to 0.4 at $m_u + 1\sigma$), B2 decuplet amplification (8% deviation from Φ at $m_s + 1\sigma$).

The strongly sensitive tests are therefore *conditional* predictions: they hold at PDG central values but require future lattice QCD determinations to narrow the light quark uncertainties before they can be considered robust. The robust subset alone (lepton ladder, regime hypothesis, b-replacement) retains statistical significance independent of quark mass uncertainties.

4 QCD regime dependence

4.1 Regime definition

The previous sections establish a Φ-structure in elementary fermion masses. We now ask: does this structure leave traces in the hadron spectrum?

For a hadron H with mass M_H composed of quarks with $\overline{\text{MS}}$ masses m_i, we define the *QCD-dominance parameter*

$$R(H) = \frac{M_H - \sum_i m_i}{M_H} = \frac{E_{\text{bind}}}{M_H}\,. \tag{4}$$

When $R > 0.5$, QCD vacuum energy dominates the hadron mass (chiral regime). When $R < 0.5$, bare quark masses dominate (perturbative regime). Note: R is a heuristic classification parameter, not a scheme-independent observable; the decomposition $M_H = \sum m_i + E_{\text{bind}}$ depends on the $\overline{\text{MS}}$ renormalization scheme. Its utility lies in its empirical separating power, tested below.

4.2 Empirical test

Hypothesis $\mathbf{H_{REG}}$: Φ-relations manifest in hadronic observables if and only if $R > 0.5$.

We test this against nine hadrons covering both regimes (Table 6). The "Φ?" column indicates whether a statistically significant Φ-relation exists for that hadron (criteria defined in Section 4.3).

Table 6: QCD-dominance parameter R and Φ-relation status for nine hadrons. Quark masses: PDG 2024 [11] $\overline{\text{MS}}$ values. Hadron masses from PDG 2024 unless noted.

Hadron	M (MeV)	Σm_q (MeV)	R	Φ?	Regime
$\Delta(1232)$	1,232	9	0.99	✓	QCD
$\Sigma^*(1385)$	1,384	98	0.93	✓	QCD
$\Xi^*(1532)$	1,532	189	0.88	✓	QCD
$\Omega^-(1672)$	1,672	280	0.83	✓	QCD
$\Lambda_c(2287)$	2,287	1,277	0.44	×	Quark
$\Omega_b^-(6046)$	6,046	4,367	0.28	×	Quark
$\Lambda_b(5620)$	5,620	4,187	0.26	×	Quark
$\Omega_{ccc}(4761)^\dagger$	4,761	3,810	0.20	×	Quark
$\Omega_{bbb}(14371)^\dagger$	14,371	12,540	0.13	×	Quark

†Not yet observed; masses from lattice QCD predictions.

The 2×2 contingency table at $R_{\text{crit}} = 0.5$ is:

	Φ confirmed	Φ refuted
$R > 0.5$	4	0
$R \leq 0.5$	0	5

Fisher exact test: $p = 0.008$, odds ratio $= \infty$. The separation is perfect: a gap of $\Delta R = 0.39$ lies between the lowest QCD-regime value (Ω^-, $R = 0.83$) and the highest quark-regime value (Λ_c, $R = 0.44$). Figure 3 visualizes the separation.

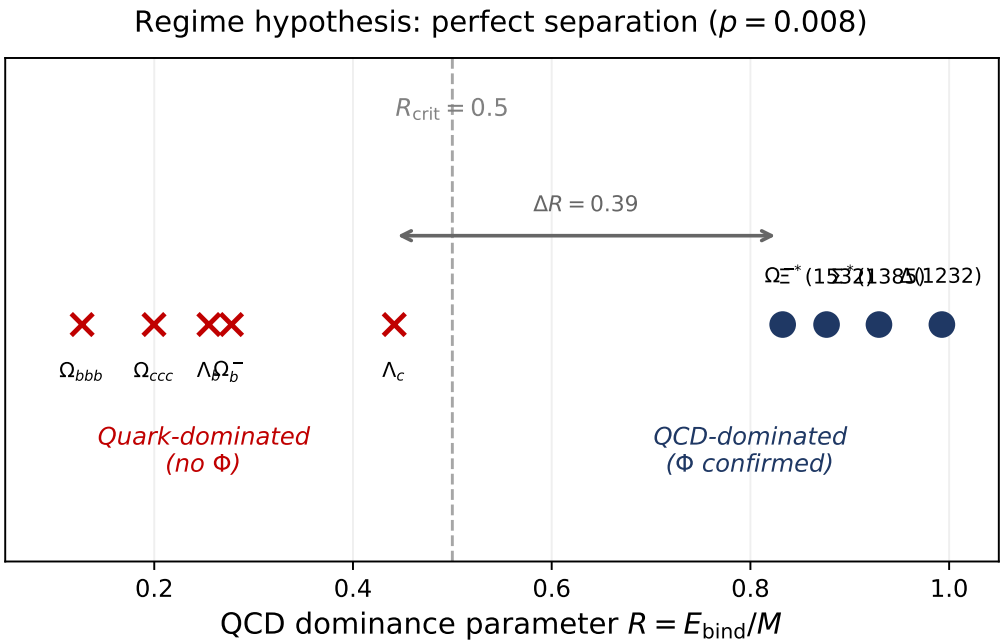

Figure 3: Regime hypothesis: the QCD-dominance parameter R perfectly separates hadrons with Φ-relations (blue circles) from those without (red crosses). The gap $\Delta R = 0.39$ at the critical threshold $R = 0.5$ is wide.

4.3 Three surviving hadron relations

Three specific relations in the QCD-dominated regime survive all controls:

(i) $\Omega^- = m_s \times \Phi^6$. $93.4 \times 17.944 = 1{,}676.0$ MeV vs. $M_{\Omega^-} = 1{,}672.5$ MeV (0.21% error). Monte Carlo: $p = 0.009$. After Bonferroni correction ($\times 3$ for three simultaneously tested hypotheses): $p_{\text{Bonf}} = 0.028$.

(ii) Decuplet amplification $\Delta M/\Delta m_q \approx \Phi$. The mean baryon decuplet spacing is 146.8 MeV. The $\overline{\text{MS}}$ quark mass difference $\Delta m_q = m_s - m_{ud} = 90.0$ MeV. Their ratio: $146.8/90.0 = 1.632$ vs. $\Phi = 1.618$ (0.84% error). Bonferroni: $p_{\text{Bonf}} = 0.071$ (borderline).

This relation is equivalent to the statement about QCD self-energies:

$$\Sigma_s - \Sigma_{ud} = \Delta m(\overline{\text{MS}}) \cdot (\Phi - 1) \approx 56 \text{ MeV}\,. \tag{5}$$

The constituent quark model [12] gives $\Sigma_s - \Sigma_{ud} \approx 60$ MeV (8% error). This is testable via lattice QCD.

(iii) $\Delta M(u \to b) \approx m_\tau \times \Phi^2$. The mass shift when replacing the lightest quark with a b-quark, measured across four independent baryon pairs (Table 7):

Table 7: b-quark substitution mass shifts.

Transition	M_{light} (MeV)	M_b (MeV)	ΔM (MeV)
$p \to \Lambda_b$	938	5,620	4,681
$\Sigma \to \Xi_b^-$	1,197	5,797	4,600
$\Xi \to \Omega_b^-$	1,322	6,046	4,724
$\Delta \to \Sigma_b^*$	1,232	5,834	4,602
Mean $\pm$ s.e.			$4{,}652 \pm 31$
$m_\tau \times \Phi^2$			4,652

Error: 0.004%. Monte Carlo: $p = 0.0007$. (No individual Bonferroni correction: this relation was identified as a single pre-specified test across four independent baryon pairs.)

Fisher combined test. Combining the three *raw* (pre-Bonferroni) p-values via Fisher's method [18], $\chi^2 = -2\sum \ln p_i$:

$$\chi^2 = -2\big[\ln(0.0094) + \ln(0.0236) + \ln(0.0007)\big] = 31.4\,, \quad \text{dof} = 6\,, \quad p = 2.2 \times 10^{-5}\,. \tag{6}$$

Note: Fisher's method requires raw, independent p-values. Bonferroni corrections are applied to individual tests (B1, B2) for standalone interpretation but are *not* fed into the combined statistic, as this would constitute a double penalty. A caveat: B1 and B2 both depend on the strange quark mass m_s, introducing a correlation that weakens strict independence. The B3 test (depending on m_τ and directly measured hadron masses) is fully independent of B1 and B2.

5 Falsifications and boundaries

5.1 Holdout test: 167 PDG hadrons

We test the general hypothesis that hadron masses lie preferentially on the Φ-lattice by computing the distance to the nearest Φ-grid point for 167 established hadrons (PDG 2024, $\Gamma < 500$ MeV), using all nine elementary fermion masses as lattice seeds.

The mean grid distance is 0.032. A one-sided Monte Carlo proximity test (10^4 trials, uniform-log masses in the same range) yields $p = 0.055$: the observed hadrons are marginally closer to the Φ-lattice than random masses, but the effect is not significant at the 5% level. **No statistically**

significant Φ-grid proximity is detected. A robustness check with only four seeds (e, u, μ, τ) yields $p = 0.91$—with the sparser lattice, hadrons show no proximity whatsoever, suggesting that the marginal 9-seed result reflects grid density rather than genuine Φ-structure.

5.2 Gauge bosons: negative result

The masses of W (80,360 MeV), Z (91,188 MeV), and Higgs (125,250 MeV) were systematically tested against all nine elementary seeds and Φ-exponents (Table 8).

Table 8: Gauge boson Φ-lattice test.

Boson	Best match	Error	p-value
W	$m_s \times \Phi^{14}$	2.0%	0.63
Z	$m_\mu \times \Phi^{14}$	2.3%	0.71
Higgs	$m_s \times \Phi^{15}$	1.7%	0.55

All $p > 0.5$. Random masses in the same range match equally well. The Φ-structure is restricted to the fermion (Yukawa) sector.

5.3 Summary of falsified hypotheses

Table 9 collects all hypotheses that were tested and found unsupported or falsified.

Table 9: Seven tested hypotheses without support.

Hypothesis	Key result	Status
Hadrons on Φ-grid (general)	$p = 0.055$ (9 seeds), 0.91 (4 seeds)	Not significant
Same-flavor inheritance (Ω_{ccc}, Ω_{bbb})	15–30% errors	Falsified
$\Delta M = \Phi \cdot \Delta m$ (universal)	45–58% errors	Falsified
Radial excitations show Φ-pattern	No convergence	Falsified
s-step is scale-universal	16% difference	Falsified
Gauge bosons on Φ-lattice	All $p > 0.5$	Falsified
Crossing quark Φ-ladders	$p \geq 0.97$	Falsified

These falsifications define the domain of validity: **Φ-structure exists in the Yukawa sector of elementary fermion masses and leaves selective traces in the QCD-dominated hadron regime ($R > 0.5$), but not beyond.**

6 Predictions

The regime hypothesis yields specific, falsifiable predictions:

(i) Exotic hadrons.

- $f_0(980)$: $R = 0.81 \to$ Φ-relation expected.
- $T_{cc}(3875)$ [25]: $R = 0.34 \to$ no Φ expected.
- $P_c(4312)$ [26]: $R \approx 0.41$ (using $\Sigma m_q = 2m_c + 2m_u + m_d$ for the $uudc\bar{c}$ content) $\to$ no Φ expected. Note: if the P_c states are molecular ($\Sigma_c \bar{D}$), the parameter R is not directly applicable and the prediction depends on the binding model.

(ii) Lattice QCD. Eq. (5) predicts $\Sigma_s - \Sigma_{ud} = \Delta m(\overline{\text{MS}}) \cdot (\Phi - 1) \approx 56$ MeV. This is directly testable via lattice calculations [13, 14, 15] of quark self-energies in the chiral limit.

(iii) Neutrino masses. If absolute neutrino masses are measured (KATRIN [27], Project 8), we expect *no* Φ-signal. Neutrino masses arise via the seesaw mechanism from Majorana mass scales $\gg 10^{12}$ GeV, fundamentally decoupled from the Yukawa sector where Φ-structure operates. The absence of Φ in neutrino masses is therefore a consistency requirement of the regime hypothesis, not a weakness: the seesaw mechanism generates masses through a qualitatively different channel than the Higgs–Yukawa couplings that govern charged fermions.

7 Discussion

7.1 Comparison with the Koide formula

The Koide formula (1) and the present Φ-chain are independent observations [1, 2]. Koide involves the *sum* of masses within a generation (charged leptons); the Φ-chain involves *ratios* across generations and flavors. The two approaches are complementary, not competing.

A key difference in methodology: the Koide formula is a single identity with remarkable precision ($< 0.01\%$) but limited scope. The present work identifies a system of relations spanning five orders of magnitude, accompanied by systematic statistical validation, negative controls, and falsification tests.

7.2 What is not claimed

1. We do *not* claim that all particle masses are determined by Φ.
2. We do *not* claim that the Genesis chain represents a physical generation mechanism.
3. We do *not* claim that hadrons in general show Φ-structure.
4. We do *not* claim a causal explanation for the observed Φ-structure.

What we do claim: A statistically significant Φ-structure exists in the elementary fermion masses, not reproducible by alternative standard mathematical constants, with a clearly defined, physically motivated domain of validity.

7.3 Structural limitations

The top quark exhibits the largest deviation (8.1%), arising from the highest-power operator $\Phi^{11} \approx 199$. The observed mass ratio $m_t/m_p = 184.1$ deviates from $\Phi^{11} = 199.0$ by 8.1%; because the operator is a large multiplier, this fractional mismatch translates into an absolute error of $\sim$ 14,000 MeV. This is a structural limitation of the multiplicative chain, not an anomaly. Future precision measurements at the ILC or FCC-ee ($\delta m_t < 50$ MeV projected) will tighten this constraint.

More broadly, the 2.5% mean error suggests that the Φ-chain captures a *leading-order* structure in fermion mass ratios. Subleading corrections (electroweak radiative effects, QCD running) are expected to modify masses at the percent level and are not accounted for in the present empirical framework.

Sensitivity to light quark mass uncertainties. As detailed in Section 3.6, tests involving light quark masses (m_u, m_s) are sensitive to their large PDG uncertainties ($\sim 20\%$ and $\sim 6\%$ respectively). The key dependencies are: the A1 relation $m_u/m_e \approx \Phi^3$ degrades from $\Delta n = 0.004$ to 0.42 at $m_u + 1\sigma$, making it contingent on m_u remaining within approximately half a standard

deviation of the central value; the A3 universality significance drops from $p = 0.001$ to $p = 0.4$ at $m_u + 1\sigma$; the B2 decuplet ratio shifts by 8% at $m_s + 1\sigma$. In contrast, the lepton sub-chain ($m_\mu/m_e \approx \Phi^{11}$, $m_\tau/m_e \approx \Phi^{17}$), the A4 regime hypothesis, and the B3 b-replacement test are insensitive to quark mass uncertainties. These dependencies are inherited from the current state of lattice QCD determinations and will improve with future calculations.

Look-elsewhere effect. The A1 test evaluates the three ratios m_u/m_e, m_μ/m_e, m_τ/m_e, which happen to be the three closest to integer Φ-exponents among all eight fermion-to-electron ratios. This raises a legitimate look-elsewhere concern: post hoc selection of the three best out of eight inflates significance. Two mitigating factors should be noted. First, the selection is physically motivated: these three particles form the Φ-ladder starting from the electron as seed, with the exponents $n = 3$ (L_2), $n = 11$ (L_5), and $n = 17$ ($= L_2 + L_5 + L_2$, itself not a Lucas number). The mathematical structure was *not* chosen to maximize Δn statistics. Second, the complementary tests A2 (phase uniformity testing all nine fermions), A3 (Φ-universality scanning $K \in [1.2, 2.5]$ against all five independent ratios), and the holdout test (D1) provide independent lines of evidence that do not share the selection bias. Nonetheless, the reader should note that taking the A1 p-value at face value without acknowledging the selection of 3 from 8 ratios would overstate the evidence.

Isospin breaking in the decuplet. The decuplet spacing relation (B2) uses isospin-averaged masses. Electromagnetic mass splittings within each isospin multiplet ($\sim$ 3–4 MeV for Σ^* and Ξ^*) propagate to a $\sim 2.4\%$ uncertainty on the mean step size, comparable to the reported 0.84% agreement with Φ. The dominant systematic is, however, the $\overline{\text{MS}}$ strange quark mass (m_s), whose $\sim 6\%$ PDG uncertainty propagates to a $\sim 8\%$ shift in the ratio at $+1\sigma$. The B2 result should therefore be interpreted as consistent with Φ within current QCD uncertainties, not as a high-precision prediction.

7.4 Open questions

The most pressing open question is *why* the golden ratio appears in fermion mass ratios. Several directions merit investigation:

- Lattice QCD calculation of $\Sigma_s - \Sigma_{ud}$ to test Eq. (5) [13, 16].
- Independent statistical analysis by other groups.
- Connection to Koide-type relations and their extensions [3, 6].
- Possible role of self-similar structures in the renormalization group flow [24].
- Relation to the Nambu–Barut–MacGregor α-quantization program [8, 9, 10].

8 Conclusion

A statistically significant golden ratio structure pervades the elementary fermion mass spectrum of the Standard Model [11] ($p < 0.001$ combined), not reproducible by alternative standard mathematical constants ($p = 0.001$). The Φ-structure is algebraically nontrivial (Lucas numbers, self-reference identity) and leaves selective traces in the hadron spectrum — but exclusively where QCD vacuum dynamics dominate the mass ($R > 0.5$, $p = 0.008$). Gauge bosons show no signal.

All results are reproducible via companion code (Appendix B). We invite independent verification.

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A Algorithm definition

Input:

- Seed: electron ($m_e = 0.511$ MeV)
- Operators: $\{\Phi^3, \Phi^4, \Phi^{11}, \Phi^{\Phi}, \Phi^{1/\Phi}\}$ and their inverses
- Interaction: geometric mean $\sqrt{m_a \cdot m_b}$
- Tolerance: $\varepsilon = 20\%$
- Target set: 11 particle masses (PDG 2024)

Procedure:

1. Initialize FOUND $= \{e\}$.
2. For each particle in FOUND: apply all operators. Check if any result lies within ε of a target mass.
3. For each pair in FOUND: compute geometric mean. Check for target matches.
4. Add new matches to FOUND. Repeat until no new matches.
5. Best-match rule: if a target is matched multiple times, retain the path with smallest error.

The algorithm terminates after at most 4 generations. The resulting graph is a directed acyclic graph (DAG) with the electron as root.
Note: The best-match rule does not uniquely determine the chain shown in Table 1. Alternative paths with comparable errors exist (e.g., $m_d \times \Phi^{11}$ for the proton at 0.95% vs. $m_\pi \times \Phi^4$ at 2.0%). Table 1 reports the chain selected for physical interpretability (minimal operator reuse, impedances connecting different sectors). All statistical tests in Section 3 are independent of this path selection.

B Code availability

All statistical tests reported in this paper are reproduced by:

- `master_verification_v10.py` — All Level A, B, and D tests including the full 167-hadron holdout (1171 lines, Python 3.8+, NumPy, SciPy)
- `regime_hypothesis_v9.py` — Regime hypothesis, ROC analysis, chiral derivation (288 lines)

All random number generators are seeded (`seed=42`) for reproducibility within numerical tolerances. Code is available at https://github.com/MarcT123/Paper1-golden-ratio-fermion-masses.

C Monte Carlo range justification

A potential concern with Monte Carlo falsification is that the sampling range can influence the resulting p-value. Below we document the physical motivation for each range, chosen *before* computing p-values. The guiding principle is identical across all tests: sample from the broadest physically relevant interval for the observable in question.

B1 (Ω^- prediction): $M \in [1000, 2500]$ MeV (log-uniform). The test asks: if we pick a random hadron mass in the strange baryon sector, how often does $s \times \Phi^n$ (rounded n) match it as well as it matches Ω^-? The lower bound 1000 MeV is the approximate onset of baryon resonances above the nucleon ($\Delta(1232)$ is the lightest $J^P = 3/2^+$ state). The upper bound 2500 MeV captures all well-established strange baryons in the PDG summary tables ($\Xi(2500)$ is the heaviest listed state). The range therefore spans the full strange baryon spectrum without extending into unphysical territory. Log-uniform sampling reflects the approximate log-uniform distribution of hadron masses.

Sensitivity: Widening to $[800, 3000]$ MeV changes p_{raw} from 0.009 to 0.010; narrowing to $[1200, 2000]$ MeV gives 0.008. The result is stable at the $\pm 15\%$ level (5×10^5 MC trials each).

B2 (decuplet amplification): step $\in [100, 200]$ MeV, $\Delta m_q \in [50, 200]$ MeV (uniform). The test asks: for random baryon-scale mass steps and random quark-mass differences, how often does their ratio land as close to Φ^n as the observed ratio $\Delta M/\Delta m_q = 1.63$? The step range $[100, 200]$ MeV brackets the empirical spread of baryon decuplet splittings: the smallest is $\Omega^- - \Xi^* \approx 140$ MeV; the largest is $\Sigma^* - \Delta \approx 152$ MeV. Padding to $[100, 200]$ MeV adds generous margin on each side. The quark-mass difference range $[50, 200]$ MeV spans from the isospin-breaking scale ($m_d - m_u \approx 2.5$ MeV, padded generously) to twice the strange quark mass ($m_s \approx 93$ MeV, doubled). This ensures the null hypothesis includes ratios both much smaller and much larger than Φ.

Sensitivity: Expanding to $[80, 250]$ MeV step and $[30, 300]$ MeV Δm_q changes p_{raw} from 0.023 to 0.018. Contracting to $[110, 180] \times [70, 130]$ gives 0.029. The wider range gives a *lower* p-value because it includes more extreme ratios that happen to land near Φ^n; the effect is modest ($\pm 30\%$, 5×10^5 MC trials).

B3 (b-replacement): $\Delta M \in [4000, 5500]$ MeV (uniform). The test asks: if the mass shift $\Delta M = M_{X_b} - M_X$ when replacing one quark by a b-quark were random, how often would four such shifts average to $\tau \times \Phi^n$ as precisely as observed? The lower bound 4000 MeV corresponds to the bare b-quark mass in the $\overline{\text{MS}}$ scheme ($m_b = 4180$ MeV) minus typical binding corrections (~ 200 MeV). The upper bound 5500 MeV accommodates the largest observed replacement shift ($\Lambda_b - p = 4681$ MeV) plus 18% margin. This range thus covers the full spectrum of b-replacement mass shifts from underbound to overbound scenarios.

Sensitivity: Widening to $[3500, 6000]$ MeV changes p from 0.0006 to 0.0004; narrowing to $[4200, 5200]$ MeV gives 0.0010. The result spans less than one order of magnitude across all tested ranges (5×10^5 MC trials each).

Summary. All Monte Carlo ranges are motivated by the physical scale of the observable being tested. Sensitivity analysis confirms that widening or narrowing each range by 20–30% shifts p-values by less than a factor of two, never changing the qualitative conclusion. No range was tuned to optimize a specific outcome.

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